

SafeRoute Kolkata

Design, Product & Technical Architecture Plan — v7

Native Kotlin Android • OpenAPI contract • device-first SOS • safety-data strategy • scaling & capacity plan • data continuity • Claude Code → GitHub → Notion → Google Cloud

Revision note (v6 → v7). v7 adds (1) a Scaling & Capacity Plan (§14), (2) Data Continuity, Backups & Excel Export (§15), and (3) a complete development workflow for **Claude Code running in Android Studio** (§17-§19): prompt 001 creates the GitHub repository and pushes to main; every later prompt runs on its own branch named after its core work, ends with a detailed commit and a pull request that Rahul reviews and merges, writes full workflow/result documentation to Notion, and draws any explanatory diagram in Excalidraw. All references to Codex are replaced by Claude Code. §0 tells the prompt-writing model (Opus, in a separate chat) how to turn this plan into prompts. Earlier change history: §2.

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0. How to Use This Document (for the Prompt-Writing Model)

Audience: this section is addressed to the model (Claude Opus, in a separate chat) that reads this PDF and writes implementation prompts. Rahul pastes each generated prompt into **Claude Code running in Android Studio's terminal** at the repository root. Claude Code has the repository, CLAUDE.md, the committed copy of this plan (docs/plan/), git, the GitHub CLI and the Notion MCP server — but it does not see this chat.

- **One prompt per request.** Generate the next prompt from the catalog (§18) unless Rahul names another. Never merge milestones or catalog entries. If an entry is too large for one reviewable pull request (rule of thumb: > ~800 changed lines excluding generated code), split it into NNNa / NNNb and say why.
- **Use the template in §19 exactly**, with every field filled. Put the branch name, commit title and PR title at the top so Rahul can check them at a glance.
- **Make the prompt self-contained.** Copy the relevant specification from this plan into the prompt (endpoints, schemas, state machines, failure-matrix rows, retention rules) and cite section numbers (e.g. 'Plan v7 §7.3'). Do not rely on Claude Code re-reading the whole PDF.
- **Respect dependencies** listed in §18. The prompt's first step is always the preflight in §17.4.
- **Workflow duties are mandatory in every prompt from P002 onwards:** own branch, quality gate, detailed commit, push, PR, repository prompt log, Notion Prompt Log page, Excalidraw diagram when §17.7 requires one, and the final report. Claude Code must never push to main and never merge a pull request.
- **Never invent secrets or IDs.** Use placeholders such as <GCP_PROJECT_ID> or <NOTION_PARENT_URL> and list them under 'Inputs Rahul must provide'.
- **Android beginner support.** Rahul has no prior Android experience. Android prompts must ask Claude Code to add a 'Learning notes' section to the Notion page explaining the Android concepts used (e.g. ViewModel, Hilt, foreground services) and to add short KDoc comments where the code is non-obvious.
- **Emergency code.** Any prompt touching SOS, live location or contacts must include the relevant rows of the SOS failure matrix (§7.5) as explicit tests or documented manual test scripts.
- **Scope discipline.** State non-goals. Claude Code must not redesign unrelated modules; if it finds a problem outside scope it records it as a follow-up in Notion instead of fixing it.
- If this plan is ambiguous for the prompt at hand, write the prompt with an explicit assumption and flag it in 'Open questions for Rahul' rather than guessing silently.

1. Product Vision & Boundaries

- Navigation-first Android app for Kolkata: map, search, walking/driving routes, safety context along routes, SOS and temporary live-location sharing.
- MapLibre + OpenStreetMap data. Safety information is **context**, never a guarantee that a route or area is safe. The UI must say so wherever a safety comparison is shown.
- **SafeRoute is not an emergency service.** Every SOS surface offers a one-tap call to 112 (India's Emergency Response Support System). The app supplements official emergency services; it never replaces them.
- No public database of alleged criminals, no naming or photographing of individuals in reports, no facial recognition, no AI crime prediction, no invented statistics, no nationwide scope in the initial product.
- No descriptions of people by religion, caste, community or appearance in any public safety content. Safety content describes *incidents and conditions* (harassment, snatching, poor lighting, deserted stretch), never *kinds of people*.

- Emergency automation (server SMS, WhatsApp, telephony, police alerting) is added only after legal, provider and operational review, and always with a fallback that works without the server.
- Launch languages: English and Bengali UI strings (Hindi post-MVP).

2. Review Findings and Change History

This section records the review of v5 and what changed. Severity: **Critical** = blocks a safe or working product; **High** = likely launch blocker; **Medium** = quality/cost/scope issue.

#	Finding in v5	Severity	Resolution in v6
F1	tRPC backend with a Kotlin client: tRPC's end-to-end types only work TypeScript-to-TypeScript. A Kotlin client would hand-code tRPC's wire format with no generated types.	Critical	OpenAPI-first HTTP contract generated from Zod schemas; Kotlin client generated from the spec (§6).
F2	SOS depends on the server being reachable ('server-side state is authoritative'). No offline path.	Critical	Device-first SOS: local countdown, device SMS, 112 dialer, foreground location; server orchestration layered on top and synced when online (§7).
F3	No Android background-execution design: foreground services, Android 14+ FGS types, OEM battery killers common in India.	High	Location-type foreground service started from a user action; no background-location permission in MVP; battery-optimisation guidance and OEM device test matrix (§5, §7).
F4	Safety data source undefined; Kolkata geocoded incident data is not publicly available.	High	Two-source strategy: official aggregates (online FIRs / RTI / partnership) at police-station level + moderated geotagged community reports at hexagon level (§9).
F5	No India-specific compliance: DPDP Act, TRAI DLT for server SMS/OTP, WhatsApp Business verification, Google Play restricted permissions and testing rules.	High	Firebase phone auth for OTP; device SMS instead of server SMS in MVP; Play declarations and closed-test plan; DPDP consent/retention/rights (§12).
F6	Map tiles, geocoding and routing providers not chosen; OSM public tile servers forbid app traffic.	Medium	MapTiler (tiles + geocoding) for MVP, self-hosted OSRM on a Kolkata extract; provider adapters kept so these can be swapped (§4).
F7	Queue, Redis and WebSocket gateway add infrastructure a solo developer must operate from day one.	Medium	pg-boss (Postgres-backed jobs) for the outbox; no Redis in MVP; live location via HTTPS + polling viewer; WebSocket gateway deferred until measured (§8, §13).
F8	Cloud Run defaults: request-scoped CPU throttles background workers; scale-to-zero cold starts on SOS paths; WebSocket connections capped by request timeout.	Medium	Worker uses instance-based billing (CPU always allocated) with min instances = 1; API min instances = 1 (§13).
F9	Scope: 23 phases designed for 1M users before any user exists; high risk of never launching.	Medium	MVP milestones M0-M10 with explicit non-goals; v5 phases moved to post-MVP roadmap (§16, §21).
F10	Live-location viewers are emergency contacts who usually will not have the app.	Medium	Viewer is a lightweight web page opened from the SMS/push link (§8).
F11	Document hygiene: v5 pages labelled v4; unresolved citation placeholders; an unverified claim about a Codex Google Cloud plugin.	Low	Consistent version labels; real references (§23); plugin claim dropped (workflow now uses Claude Code).

2.1 Change history

Version	Main changes
v4	tRPC API architecture (replaced REST).
v5	Codex → GitHub → Google Cloud deployment path, Workload Identity Federation, CI/CD.
v6	Kotlin native kept; tRPC → OpenAPI; device-first SOS; simplified live location; safety-data strategy; India compliance; MVP milestones; review findings F1-F11 above.
v7	Scaling & capacity plan (§14); data continuity, backups & Excel export (§15); Claude Code workflow in Android Studio with branch-per-prompt, PR review, Notion documentation and Excalidraw diagrams (§17-§19); prompt-writing guide (§0); reports schema extended for long-term data use (§11).

3. MVP Definition

3.1 Goal

Launch a reliable, privacy-respecting Android app in Kolkata that lets a person (a) plan a route and see transparent community/official safety context along it, (b) share live location temporarily with trusted contacts, and (c) trigger an SOS that reaches their contacts even when the SafeRoute server or mobile data is unavailable. Launch in a limited set of pilot areas first, then widen coverage as report data grows.

3.2 In scope (MVP features)

ID	Feature	MVP behaviour
F-01	Sign-in	Phone number + OTP via Firebase Authentication; backend verifies Firebase ID tokens. Onboarding with DPDP consent notice (English/Bengali).
F-02	Map	MapLibre Android map, MapTiler style, current location with accuracy circle.
F-03	Search	Place/address search via backend geocoding proxy (API key never shipped for geocoding).
F-04	Routes	Walking and driving routes via OSRM with up to 3 alternatives, drawn on map; follow-me mode with live position. No voice turn-by-turn in MVP.
F-05	Safety overlay	Hexagon cells showing counts of approved community reports (with time window and category filter) + official area aggregates when available. Always labelled with source and time window.
F-06	Route comparison	Each alternative shows its exposure metric v1 (§10) as counts, not a score.
F-07	Emergency contacts	Up to 5 contacts; each is informed when added (invite SMS from the user's phone or push if they have the app) and can opt out.
F-08	SOS	Hold-to-arm + 5 s cancellable countdown → device SMS to contacts, 112 dialer, foreground location tracking, server session + push to contacts with the app, 'I'm safe' follow-up.
F-09	Live share	Temporary sharing (15 min / 1 h / 8 h / until stopped) with revocation; web viewer.
F-10	Report incident	Geotagged report with fixed categories and time-of-incident; moderated before it counts publicly.
F-11	Moderation	Minimal internal web page for moderators (approve/reject/redact) with audit log.
F-12	Account & data	In-app and web account deletion; data export on request; retention jobs.

3.3 Explicit non-goals for the MVP

- iOS app; Wear OS; voice turn-by-turn navigation; offline map packs.
- Server-sent SMS, WhatsApp messages, automated phone calls, automatic police-station alerting.
- Redis, separate message broker, WebSocket gateway, read replicas, microservices.
- Public free-text or photos in reports; comments; upvotes; user profiles visible to others.

- Any 'safety score', prediction, or ranking of neighbourhoods.

3.4 Launch criteria

- All SOS failure-matrix tests in §7.5 pass on at least one Pixel/stock device and two OEM devices with aggressive battery management (e.g. Xiaomi/Redmi and Vivo/Oppo/Realme).
- Closed test with ≥ 12 opted-in testers for 14 consecutive days completed (required for new personal Play developer accounts) and crash-free sessions $\geq 99\%$ in that period.
- Play Console: Data safety form, privacy policy URL, account-deletion URL, foreground-service declaration, SEND_SMS permissions declaration (or SMS-intent-only build), content rating, target API level 36.
- Moderation SLA defined (e.g. reports reviewed within 48 h) and at least one backup moderator.
- Restore-from-backup drill completed for Cloud SQL; rollback of API revision rehearsed.

3.5 Success metrics (first 90 days)

- Activation: % of new users who add ≥ 1 emergency contact.
- Reliability: % of SOS triggers where ≥ 1 contact alert (SMS or push) is confirmed sent within 30 s; target $\geq 99\%$.
- Live share: median location staleness seen by viewers during active sessions; target < 20 s.
- Data: approved reports per week in pilot areas; moderation turnaround time; rejection rate.

4. Core Architecture (MVP Topology)

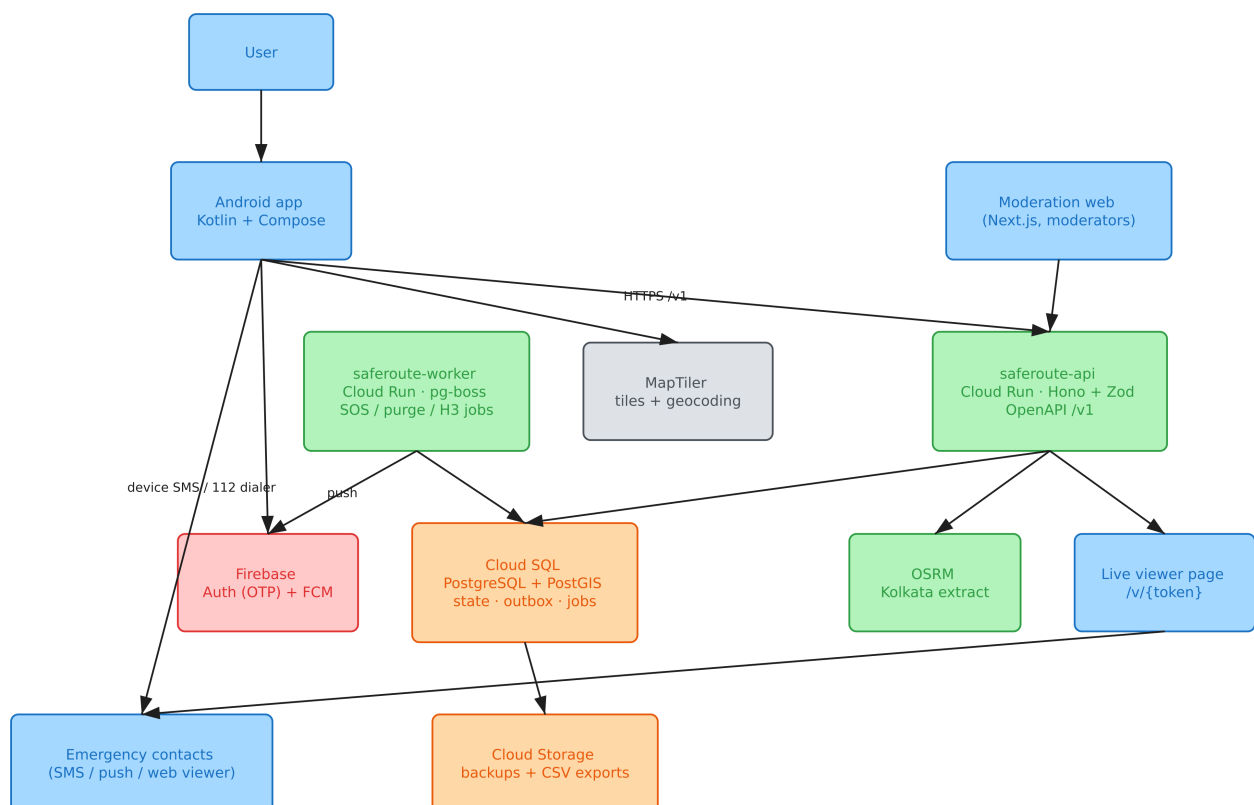


Diagram source: docs/diagrams/02-mvp-architecture.excalidraw (open at excalidraw.com to edit).

- Modular monolith:** one API codebase organised by domain modules (auth, users, contacts, search, routing, safety, reports, sos, share, admin). Extract services only when measurements justify it.

- **PostgreSQL/PostGIS is the only durable store** in the MVP. Jobs use pg-boss, so the outbox row and the job enqueue can commit in the same transaction.
- **SOS isolation**: SOS jobs run on a dedicated pg-boss queue with its own concurrency; retention/aggregation jobs can never starve SOS work.
- **Provider adapters** (tiles, geocoding, routing, push, later SMS/WhatsApp/telephony) sit behind interfaces in the backend so providers can be replaced without touching the app contract.

5. Android Client

5.1 Stack

Concern	Choice	Notes
Language / UI	Kotlin, Jetpack Compose, Material 3	Single-activity app; Navigation Compose.
DI	Hilt	
Networking	Retrofit + OkHttp + kotlinx.serialization	Client code generated from openapi.json; auth interceptor attaches Firebase ID token.
Async	Coroutines + Flow	
Local storage	Room + DataStore	Room holds contacts, pending SOS/sync queue, last known location — everything SOS needs offline.
Background work	WorkManager + foreground service (type location)	WorkManager for retryable sync; FGS for SOS and live share.
Location	Fused Location Provider (Google Play services)	High accuracy during SOS; balanced otherwise.
Maps	MapLibre Native Android SDK	MapTiler style URL; attribution for OSM and MapTiler shown.
Auth	Firebase Authentication (phone)	Firebase handles OTP SMS delivery.
Push / crashes	Firebase Cloud Messaging, Crashlytics	No precise location or message text in crash logs.
Build	Gradle Kotlin DSL + version catalog	minSdk 26, targetSdk 36 (required for Play submissions since 31 Aug 2026).
Tests	JUnit, Turbine, Compose UI tests, Robolectric where useful	

5.2 Project structure

Start as a **single Gradle module with feature packages** — simpler while learning Android. Split into Gradle modules (core:network, core:data, core:location, feature:*) only if build times or boundaries hurt.

```
android/app/src/main/java/in/saferoute/  
  core/      designsystem/ network/ (generated client) data/ (Room, repositories) location/ auth/  
  feature/   onboarding/ map/ search/ route/ sos/ share/ contacts/ reports/ settings/  
  service/   SosForegroundService.kt ShareForegroundService.kt  
  work/      SosSyncWorker.kt LocationUploadWorker.kt
```

5.3 Permissions (MVP)

Permission	Why	Policy notes
ACCESS_FINE_LOCATION / COARSE	Map, routing, SOS, share, reports	Requested in context with rationale.
FOREGROUND_SERVICE, FOREGROUND_SERVICE_LOCATION	SOS and live share tracking while app is minimised	Play foreground-service declaration with demo video. Service is started from a user action, so ACCESS_BACKGROUND_LOCATION is not requested in the MVP.
POST_NOTIFICATIONS	FGS notification, alerts to contacts	Android 13+ runtime permission.
SEND_SMS	Automatic SOS SMS to contacts without data connectivity	Restricted permission. Play lists emergency/physical-safety SMS alerts as an eligible use; requires a Permissions Declaration Form and approval. Until approved, ship the SMS-intent fallback (one tap).
INTERNET	API calls	

5.4 IDE and tooling

- **Android Studio** (stable) is the main IDE. Install the Android 16 (API 36) SDK platform. Android Studio opens /android as the Gradle project; Compose previews, emulator, Logcat and the profiler are used from there.
- **Claude Code** runs in Android Studio's built-in terminal **from the repository root** (not from /android), so it can see backend, contracts, docs and CI. Install the Claude Code plugin for JetBrains IDEs so diffs open in the IDE's diff viewer. See §17.1 for the full tool list.
- Backend/TypeScript files are edited by Claude Code; Rahul reviews them in the GitHub pull request (or VS Code if preferred — Android Studio's TypeScript support is limited).
- Physical test devices are required: emulators do not reproduce OEM battery killers or real GPS behaviour.
- Learning path before P007: Kotlin basics → Compose basics codelabs → the 'Now in Android' sample as a reference architecture.

6. API Architecture — OpenAPI Contract (replaces tRPC)

6.1 Why the change

tRPC shares types by importing the server's TypeScript router type into a TypeScript client. The SafeRoute client is Kotlin, so that mechanism is unavailable. An OpenAPI contract keeps the same benefit — one schema source of truth and generated, typed clients — across languages. The backend keeps TypeScript and Zod, so the developer experience on the server barely changes.

6.2 Rules

- Backend: Node.js (current LTS) + Hono + @hono/zod-openapi. Every route declares Zod request/response schemas; the OpenAPI 3.1 document is generated at build time and committed as contracts/openapi.json.
- Kotlin client generated with openapi-generator (Kotlin, Retrofit2, kotlinx.serialization) in CI; the app never hand-writes request/response models.
- Versioned path prefix /v1. CI runs a breaking-change diff (e.g. oasdiff) on openapi.json; breaking changes require an ADR and a new version path or a coordinated app release, because old app versions stay installed for months.
- Errors use RFC 9457 problem+json with a stable code field the app can switch on.

- State-changing, retryable calls (SOS, reports, share creation, location batches) accept an Idempotency-Key header; the server stores key → response for 24 h.
- Authorization is enforced in server middleware and domain services; the client is never the security boundary.
- Per-route rate limits (Postgres-backed token buckets in MVP): reports, share creation, SOS creation, public viewer polling, search.
- ORM: Drizzle ORM with PostGIS geometry columns and raw SQL for spatial queries (Prisma is acceptable only if spatial queries live in typed raw SQL and GiST indexes are managed in migrations).

6.3 MVP endpoints

Domain	Endpoints
Me	POST /v1/me/bootstrap · GET /v1/me · PATCH /v1/me · DELETE /v1/me (account deletion) · POST /v1/me/export
Devices	PUT /v1/devices/{installationId} (FCM token, app version)
Contacts	GET /v1/contacts · POST /v1/contacts · DELETE /v1/contacts/{id} · POST /v1/contacts/{id}/opt-out (public, token-based)
Search	GET /v1/search?q=&near=
Routing	POST /v1/routes {origin, destination, mode, departAt} → alternatives + exposure metrics
Safety	GET /v1/safety/cells?bbox=&window=&categories= · GET /v1/safety/areas?bbox=
Reports	POST /v1/reports · GET /v1/reports/mine · DELETE /v1/reports/{id}
SOS	POST /v1/sos (Idempotency-Key, clientSosId) · POST /v1/sos/{id}/locations · POST /v1/sos/{id}/resolve · GET /v1/sos/{id}
Share	POST /v1/shares · POST /v1/shares/{id}/locations · POST /v1/shares/{id}/stop
Public viewer	GET /v/{token} (HTML) · GET /v1/public/shares/{token}/latest
Admin	GET /v1/admin/reports?status=pending · POST /v1/admin/reports/{id}/decision · GET /v1/admin/audit

7. SOS — Device-First Design

Principle: the actions that matter most in an emergency — alerting contacts and reaching 112 — must not depend on the SafeRoute server, mobile data, or the app staying in the foreground. The server adds live tracking, push alerts, audit and orchestration on top.

7.1 Trigger flow

- **Arm:** press and hold the SOS button for 2 s (prevents pocket triggers). Later: Quick Settings tile and home-screen widget.
- **Countdown:** 5 s on-device countdown with vibration and a large Cancel button. Cancellation during the countdown is purely local — nothing is sent anywhere.
- **At expiry, in parallel on the device:**
 - Persist an SOS record in Room with a client-generated clientSosId (UUID v7).
 - Start SosForegroundService (type location): high-accuracy fixes every 5 s, stored locally.
 - Send SMS to all contacts via SmsManager if SEND_SMS is granted; otherwise open the SMS composer (ACTION_SENDTO) pre-filled with all numbers. Message: name, 'needs help', map link with coordinates, accuracy, time, and the live link if already available.
 - Show the full-screen SOS screen with a primary **Call 112** button (opens the dialer with 112 — Android routes emergency numbers through the system dialer).
 - Enqueue SosSyncWorker (WorkManager, expedited, exponential backoff) to POST /v1/sos.

7.4 Alert channels by phase

Channel	MVP	Post-MVP
SMS from the user's own phone	Yes (SmsManager or SMS intent)	Kept as the offline fallback
112 dialer	Yes, one tap	Optional auto-open setting
Push to contacts who installed the app	Yes (FCM via worker)	Rich notification with map
Server SMS (provider)	No — requires TRAI DLT sender/template registration, usually as a registered entity	Add after DLT registration
WhatsApp	No — business verification + approved templates	Add after verification
Automated voice call	No	Evaluate with legal review

7.5 Failure matrix (must pass before launch)

Scenario	Expected behaviour
No mobile data, cellular signal present	SMS sent from device; server sync queued; live link SMS sent when sync succeeds.
No signal at all	SOS screen shows 112 button; device keeps retrying; local record kept; SMS retried when signal returns.
SafeRoute API down / 5xx	Device path unaffected; WorkManager retries; no duplicate sessions (idempotency).
App swiped away / killed by OEM	Foreground service keeps running where allowed; on process restart, state restored from Room and service resumed.
Phone rebooted during SOS	On boot/app open, active SOS detected in Room and user prompted to continue or resolve.
Double tap / duplicate trigger	Single session (clientSosId); second trigger shows existing SOS screen.
SEND_SMS denied or not approved	SMS composer opens pre-filled; user taps send.
No GPS fix	Send last known location with its age; mark as stale in SMS and viewer.
FCM failure	Action recorded FAILED_RETRYABLE, retried; SMS already covers contacts.
Worker crash mid-action	pg-boss re-delivers; provider call keyed by action idempotency key; no duplicate push.
Accidental trigger	Cancel in countdown; after trigger, 'I'm safe' sends a clear follow-up to contacts.

8. Live Location Sharing (MVP)

- Session durations: 15 min, 1 h, 8 h or until stopped. Revocation is immediate; a revoked or expired link returns HTTP 410 and the viewer shows 'sharing ended'.
- Viewer capability: 128-bit random token (base64url) in the URL; only its SHA-256 hash is stored. No predictable IDs.
- Device: ShareForegroundService uploads batches every 10–15 s (5–10 s during SOS) with latitude, longitude, accuracy, device timestamp, provider, battery level (optional, user-controlled).
- Server: upserts the latest snapshot on the session row and keeps a capped trail (last ~50 points) for the viewer. Trail and snapshot are deleted 24 h after the session ends; SOS sessions follow §12 retention.
- Viewer: `/v/{token}` web page with MapLibre GL JS; polls the latest endpoint every 5–10 s; shows 'updated N s ago', accuracy circle and a stale warning after 60 s.

- Upgrade path: SSE or a WebSocket gateway only when polling load or latency requirements are measured to need it.

9. Safety Data Strategy

Two complementary sources: **official records** give credibility but are coarse, delayed and exclude the most safety-relevant categories; **geotagged community reports** give street-level, time-of-day detail and capture incidents that never reach a police station, but need strong abuse controls. Neither is shown as a 'safety score'.

9.1 Police records (thana / police chowki)

Collecting data directly from police-station complaint registers is not a workable route: FIRs and General Diary entries are police records containing personal data of complainants, victims and accused, and station staff are not in a position to hand them over. Use these legitimate channels instead:

Route	What you get	Limits / cautions
Online FIRs published under the Supreme Court's 2016 Youth Bar Association direction	Individual FIRs for cognisable offences, uploaded to police websites	Sexual offences, POCSO, insurgency and terrorism are exempt from publication — the categories most relevant to personal safety are missing. FIRs contain names and addresses: extract only offence category, date, police station and approximate place; never store personal fields. Check the website's terms before any automated collection.
RTI Act 2005 requests	Aggregated counts by police station × offence category × month (and time band if recorded)	Ask only for anonymised aggregates so the personal-information exemption does not apply. Greater Kolkata spans several police jurisdictions (Kolkata Police and neighbouring commissionerates) — file with each.
Formal partnership / MoU with Kolkata Police	Regular aggregate feeds; possible official channel for SOS escalation later	Slow; a working MVP with privacy safeguards strengthens the pitch.
NCRB 'Crime in India'	City-level annual statistics	Context only; not usable along routes.

- Granularity: official data is attached to **police-station jurisdiction polygons** (outposts/chowkis roll up to their thana) and displayed as area-level context, never as points.
- Jurisdiction boundary polygons may need to be digitised from published maps — tracked as an open item.
- Every record keeps provenance: source, reference, period covered, received date, verification method, metric version.

9.2 Geotagged community reports

Recording incidents with a geotag once the app is live is a good idea and becomes the primary street-level source. Design rules:

Aspect	Rule
Categories (fixed list)	Verbal harassment / eve-teasing · Following / stalking · Physical harassment · Snatching / theft · Poorly lit stretch · Deserted stretch · Unsafe transport stop · Other
Location	From GPS; user may move the pin within ~300 m (reporting after leaving the spot is safer). Exact point is visible only to moderators and is coarsened after moderation.
Time	Time of incident in 1-hour bands; never shown publicly to the minute.
Free text	Optional, private to moderators in MVP; moderators strip names, phone and vehicle numbers.
Media	No photos or video in MVP (identity and consent risks).

Aspect	Rule
Identity	Reporter identity never shown. The report is linked to the account only for abuse control.
Publication delay	Reports appear only after moderation and a minimum delay (e.g. 24 h), so the reporter cannot be identified by someone who saw them at the spot.
Aggregation	Displayed on H3 hexagons (resolution 9, roughly 0.1 km ²). A cell shows data only when it has reports from ≥ 3 distinct reporters in the window (k-threshold).
Time decay	Default window: last 180 days; older data fades from default views.
Abuse controls	Verified phone; account age ≥ 24 h before reporting; rate limit (e.g. 3/day); duplicate clustering; reporter trust score; moderators reject reports that describe people by religion, caste, community or appearance, or that target a named person or business.
Official path	After submitting, the app links to official channels (112, police complaint portals) — a SafeRoute report is not a police complaint, and the app says so.

- Legal notes: never publish anything that could reveal the identity of a victim of a sexual offence (a criminal offence under the Bharatiya Nyaya Sanhita); avoid defamatory content about identifiable people or businesses; treat report location and time as personal data under the DPD Act.
- Cold start: at launch the overlay will be sparse. Label it 'Community reports (beta)', launch in a few pilot areas, and add OSM context (police stations, hospitals, 24-hour fuel stations/pharmacies where tagged) so the map is useful before reports accumulate.

10. Route Exposure Metric v1

Routing and safety stay separate: OSRM returns candidate routes; the safety module computes exposure over each route's geometry. The metric is deterministic, versioned and explained in the UI.

- Buffer each route by 75 m (PostGIS ST_Buffer on a projected geometry).
- Community component: approved reports within the buffer and the selected window, deduplicated by cluster; weight = recency (half-life 90 days) $\times$ time-of-day match (1.0 within ± 2 h of departure time, 0.5 otherwise).
- Official component: police-station areas the route passes through, with their aggregate counts for the period.
- Output shown as text, e.g. '4 community reports along this route in the last 6 months (2 at this time of day); passes through 2 police-station areas'. No single score; no 'safe' label.
- Store `metric_version` with each computation; changes to weights require an ADR.

11. Data Model (MVP)

Table	Key columns / notes
users	id, firebase_uid (unique), phone_e164, display_name, locale, role, created_at, deleted_at
devices	installation_id, user_id, fcm_token, app_version, last_seen_at
emergency_contacts	id, user_id, name, phone_e164, has_app_user_id (nullable), opted_out_at, invited_at
sos_sessions	id, user_id, client_sos_id (unique per user), state, version, triggered_at_device, received_at, resolved_at, share_session_id
sos_actions	id, sos_session_id, type (push/sms/...), target_contact_id, state, attempt_count, idempotency_key, provider, provider_ref, last_error, timestamps
outbox	id, aggregate, event_type, payload (no message text), created_at, published_at — pg-boss jobs
share_sessions	id, user_id, token_hash, purpose (share/sos), expires_at, revoked_at, latest_point (geometry), latest_accuracy_m, latest_at
share_points	share_session_id, point, accuracy_m, recorded_at — capped, purged after session
reports	id, user_id, category, taxonomy_version, occurred_band, point_exact (moderator-only, 30 days), cell_h3_r11, cell_h3_r9, cell_lat, cell_lng, collection_phase (closed_test open_beta ga), status, trust_weight, created_at, published_at
report_decisions	report_id, moderator_id, decision, reason_code, redactions, decided_at
official_aggregates	id, data_source_id, area_id, category, period_start, period_end, count
data_sources / areas / police_stations	provenance records; jurisdiction polygons (geometry)
route_metrics_cache	route_hash, metric_version, window, result, computed_at (short TTL)
idempotency_keys	key, user_id, route, response_hash, expires_at
audit_log	actor, action, entity, entity_id, metadata (no precise location), at

12. Security, Privacy & Compliance

12.1 DPDP Act (India)

- Clear consent notice at onboarding in English and Bengali, stating purposes: navigation, SOS alerts, live sharing, safety reports. Separate, revocable consent for each optional purpose.
- Contacts' phone numbers are personal data of people who are not users: inform them when added and give an opt-out link.
- Rights: access/export, correction, erasure (account deletion), grievance contact published in-app and on the website. Breach-response runbook.
- The DPDP Rules have phased commencement — verify which obligations apply at launch date and record the review in an ADR.

12.2 Retention

Data	Retention
Live-share points and snapshot	Deleted 24 h after the session ends
SOS location trail	30 days (for user review/recovery), then deleted; session metadata kept 1 year for audit, without location
Report exact point	Coarsened to H3 cell after moderation; exact point deleted after 30 days
Application logs	30 days; never contain OTPs, tokens, precise locations or SMS text
Deleted accounts	Personal data erased within 30 days; approved reports kept only as anonymous aggregates

12.3 Google Play requirements checklist

- Target API level 36 for new apps and updates (since 31 Aug 2026).
- New personal developer accounts: closed test with ≥ 12 opted-in testers for 14 consecutive days before production access. (Organisation accounts are exempt but need a legal entity and D-U-N-S number.)
- Data safety form; privacy policy URL; in-app and web account deletion.
- Foreground service (location type) declaration with video; SEND_SMS permissions declaration under the emergency/physical-safety alert use case — remove SEND_SMS from the manifest if not approved.
- Prominent disclosure before requesting location.

12.4 Security controls

- Firebase ID token verification on every protected route; roles (user, moderator, admin) via custom claims checked server-side.
- Threats carried from v5: OTP abuse (Firebase App Check + quotas), fake SOS spam (per-user limits, no public amplification), location scraping (capability tokens, rate limits, no enumeration), contact takeover, duplicate actions, moderator abuse (audit log, two-person rule for bulk actions), report brigading.
- Secrets in Secret Manager; only a restricted, quota-alerted tile key ships in the app.
- App Check / Play Integrity on report and SOS endpoints (post-MVP hard enforcement, MVP monitor mode).

13. Google Cloud Deployment (MVP) and CI/CD

13.1 Topology

Component	MVP configuration
Region	asia-south1 (Mumbai) for latency to Kolkata; confirm service availability and pricing
Projects	saferoute-staging and saferoute-prod (separate projects, separate Firebase projects)
saferoute-api	Cloud Run service, min instances 1, concurrency tuned after load test
saferoute-worker	Cloud Run service with instance-based billing (CPU always allocated), min instances 1, no public ingress
osrm	Cloud Run service with the Kolkata graph baked into the image (or a small Compute Engine VM if memory/start-up time requires)
Database	Cloud SQL for PostgreSQL + PostGIS, automated backups + point-in-time recovery, private IP
Migrations	Cloud Run Job using sa-migration identity
Scheduled jobs	Cloud Scheduler → worker endpoints (purge, aggregation)
Other	Secret Manager, Artifact Registry, Cloud Logging/Monitoring, budget alerts
Moderation web	Cloud Run service behind IAP or moderator-role auth

13.2 CI/CD (kept from v5, adjusted)

- Pull request: dependency/security checks → typecheck → unit/integration tests (Postgres+PostGIS in a container) → OpenAPI generation + breaking-change diff → Kotlin client generation → Android lint/unit tests and assemble → migration validation → container build. No deployment.
- Merge to main: build immutable image tagged with commit SHA → Artifact Registry → deploy to staging → smoke tests → manual approval → deploy the same image to production → verify

error/SLO metrics → roll back to the previous revision on failure.

- Android release: signed AAB built in CI; uploaded to the Play internal/closed track; production promotion is manual.
- Authentication: GitHub OIDC → Workload Identity Federation, scoped to the SafeRoute repository and protected environments. No service-account JSON keys anywhere.
- Identities: gcp-deploy-staging, gcp-deploy-production, sa-api-runtime, sa-worker-runtime, sa-migration — each least-privilege.
- Schema changes: expand → deploy → migrate/backfill → contract. No SOS-critical release depends on an untested destructive migration.
- Claude Code never receives production credentials; it works only through branches, PRs and CI.

14. Scaling & Capacity Plan

The MVP is built so it can scale without a rewrite: the API is stateless and scales horizontally on Cloud Run; PostgreSQL is the single durable store; providers sit behind adapters; and SOS works on the device first, so an overloaded server never stops an SOS from reaching contacts. Scaling steps are added **only when measurements cross a trigger**, never in advance.

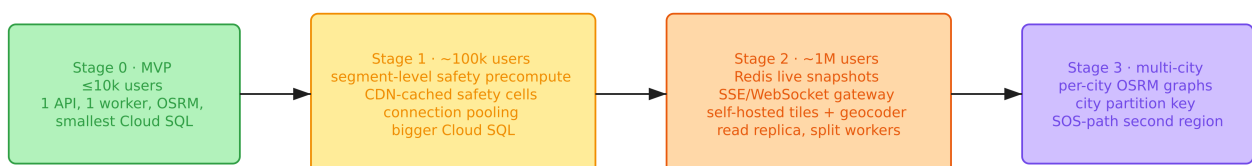
14.1 Capacity model (assumptions stated)

Assumptions for planning: 1M registered users in Kolkata; 10% daily active (100k DAU); 15% of daily traffic in the peak hour; peak bursts 3× the peak-hour average. Registered users ≠ concurrent users.

Workload	Daily volume	Peak load	Verdict / first bottleneck
General API (profile, contacts, reports, safety cells)	~20 calls/DAU → 2M/day	~100–250 req/s	A handful of Cloud Run instances.
Routing	~2 routes/DAU → 200k/day	~30–100 req/s	One OSRM instance handles a city graph; the per-route safety computation is the expensive part.
Live-location uploads	~3k share sessions/day	~35 writes/s normal; ~340 writes/s festival peak	Postgres copes initially; first workload to move out.
Live-location viewers (polling)	2 viewers/session, 5 s poll	~140 req/s normal; ~1,400 req/s festival peak	Needs caching or push (SSE/WebSocket) at scale.
SOS	~1 per 1,000 DAU/day → ~100/day	Tiny	Throughput is not the risk — isolation, burst behaviour and correctness are.
Map tiles	50–100 tiles per map session → ~15M/day	Very high	Cost problem more than load problem (per-request provider pricing).

Kolkata's design peak is Durga Puja, not an ordinary evening: very large crowds, heavy live sharing and many anxious contacts watching. Plan for 5–10× normal live-location and viewer load during Puja nights and load-test that scenario before every festival season.

14.2 Scaling stages and triggers



Stage	Add these	Trigger (measured)
0 · MVP (≤10k users)	v6 topology: 1 API, 1 worker, OSRM, smallest Cloud SQL; dashboards from day one.	—
1 · ~100k users	Precompute safety exposure per OSM road segment nightly (route metric = sum over segments instead of a PostGIS buffer query per request) · CDN-cacheable safety-cell responses keyed by H3 tile + window · small per-instance DB pools + Cloud Run max-instance cap + PgBouncer (or Cloud SQL managed pooling where available) · scale Cloud SQL vertically first.	DB CPU > 60% sustained, p95 route latency > 1.5 s, or DB connections > 70% of limit.
2 · ~1M registered / Puja peaks	Live-location snapshots in Redis (Memorystore); batch or skip point writes to Postgres · SSE/WebSocket realtime gateway fed from Redis (replaces viewer polling) · self-hosted tiles (Protomaps PMTiles on Cloud Storage + Cloud CDN) and geocoder (Photon or Pelias) · read replica for map/safety reads · separate worker pools with reserved SOS capacity · time-partitioned trail and job tables.	Viewer polling > ~500 req/s, provider bill growth, Puja load test fails SLOs, job-queue age > 30 s.
3 · multi-city	Per-city OSRM graphs (or one routing service with all graphs) · city as partition key for reports/safety · optional second region for the SOS path only.	Decision to launch a second city.

14.3 SLOs

Signal	Target
SOS: trigger → first contact alert confirmed sent (SMS or push)	p99 ≤ 30 s
Routing response incl. exposure metric	p95 ≤ 2 s
Live-location staleness seen by viewer	p95 ≤ 20 s
API availability (non-SOS)	99.5% monthly
SOS job queue age	p99 ≤ 10 s

14.4 Load testing (k6)

- Scenario A — normal evening peak (mixed API, routing, live share).
- Scenario B — Puja night: 10× live sharing and viewers, 3× routing.
- Scenario C — SOS burst: many triggers within minutes while Scenario A runs; verify SOS SLOs hold.
- Pass criteria are the SLOs above; results recorded in Notion. Run the baseline before launch (P021) and Scenario B before each festival season.

14.5 Capacity dashboard (from M1)

Request rate and p95/p99 latency per route · Cloud Run instance count · Cloud SQL CPU, memory, connections, slow queries · pg-boss queue depth and age per queue · SOS time-to-first-alert · live sessions and viewer requests · provider request counts and spend · budget alerts.

14.6 What not to do

No microservices, Kubernetes or multi-region deployment until a load test identifies a concrete limit that the steps above cannot fix. Even at Stage 3 the system should still be a modular monolith plus a realtime gateway.

15. Data Continuity, Backups & Excel Export

The beta's main long-term asset is the incident data it collects. That data lives in PostgreSQL, not in the app, so rewriting the app, replacing the backend or scaling up does not lose it — provided the rules below are followed from the first public build.

15.1 Keep the product identity stable

- Choose the Android package name once (e.g. `in.saferoute.app`); it can never change after publishing. Use Play App Signing so any future rewrite ships as an update to the same listing.
- Keep the same Firebase projects so user IDs stay stable; if auth ever moves off Firebase, export users and map IDs in a migration.
- Run the public beta against **production**, not staging.

15.2 Store data so it stays useful

- Store raw facts, not conclusions: category, time band, location cells, moderation status, provenance. Scores and layers are computed at query time and can be recomputed when methods improve.
- Store the H3 cell at a fine resolution (res 11) plus the display resolution (res 9), and the cell centre latitude/longitude. Coarse views can always be rebuilt from fine cells, never the reverse — this matters because exact points are deleted after 30 days (§12.2).
- Version the category taxonomy (taxonomy_version) so old reports can be mapped to new categories.
- Tag every report with collection_phase (closed_test | open_beta | ga) so tester and trial data can be excluded later.
- Evolve the schema only through expand → migrate → contract migrations; a 'new app' means new code against the same database, never a fresh empty one.
- Consent notice wording covers use of reports for safety context in SafeRoute *and its future versions*. Any genuinely new purpose (e.g. sharing with police or researchers) needs new consent or anonymised aggregates.

15.3 Backups

- Cloud SQL automated daily backups + point-in-time recovery (production).
- Weekly scheduled export of the database to a Cloud Storage bucket in a different region, with long retention and object versioning.
- Quarterly restore drill into a scratch instance; result recorded in Notion (runbook in docs/runbooks).

15.4 Exporting data to Excel

Method	How	Use for
DBeaver (GUI)	Run Cloud SQL Auth Proxy locally → connect DBeaver to localhost:5432 with the read-only export_readonly user → run query → Export data → XLSX.	Occasional analysis
psql \copy	With the proxy running: <code>\copy (SELECT ...) TO 'reports.csv' WITH CSV HEADER</code> ; open in Excel via Data → From Text/CSV → UTF-8.	Repeatable exports, scripts
Moderation web 'Export'	Button that runs an approved query and returns CSV/XLSX; restricted to moderators/admins; every export written to audit_log.	Long-term standard route (P018)
gcloud sql export csv	Server-side export of a query to Cloud Storage, then download.	Scheduled exports

- Export anonymised fields only by default: no phone numbers, user IDs, exact points or free text.
- Geometry is exported as numbers: `ST_Y(point)/ST_X(point)` or the stored cell centre latitude/longitude.
- Convert timestamps with `AT TIME ZONE 'Asia/Kolkata'`; open CSVs as UTF-8 so Bengali text survives; Excel's limit is ~1,048,576 rows per sheet.
- Excel files are working copies, not the archive; the database and its backups are the archive. An export on a laptop is still personal data — keep it encrypted and delete old copies.

16. MVP Milestones (M0-M10)

Durations are rough estimates for one developer learning Android while building; adjust after M2.

Milestone	Deliverables	Exit criteria	Est.
M0 Setup & learning	Android Studio, Kotlin + Compose basics, throwaway map app; monorepo scaffold (/android, /backend, /moderation, /contracts, /infra, /docs); ADR-001 (Kotlin + OpenAPI)	Throwaway app shows MapLibre map with location	1–2 wk
M1 Backend foundation	Hono API, health, Firebase token auth, Drizzle + PostGIS migrations, OpenAPI generation, CI, staging deploy via WIF	Staging /health green from CI deploy	1–2 wk
M2 Android foundation	App shell, navigation, Hilt, generated client, Firebase phone sign-in, onboarding + consent (EN/BN)	Sign-in → /v1/me works on device	2 wk
M3 Map & location	MapLibre + MapTiler, location permission flow, current location	Works on 3 test devices	1–2 wk
M4 Search & routing	Geocoding proxy, OSRM Kolkata deploy, alternatives drawn, follow-me mode	Route between two Kolkata points in < 2 s p95 on staging	2 wk
M5 Contacts + SOS (device)	Contacts CRUD + invite, arm/countdown, FGS, SMS (manager + intent), 112 dialer, Room persistence	Device-side rows of failure matrix pass offline	2–3 wk
M6 SOS (server) + live share	SOS endpoints, idempotency, pg-boss worker, FCM push, share sessions, viewer page	All failure-matrix rows pass	2–3 wk
M7 Reports + safety layer	Report flow, moderation web, H3 aggregation, overlay, exposure metric v1	Moderated report appears in cell after delay	2–3 wk
M8 Privacy & hardening	Account deletion/export, retention jobs, rate limits, Crashlytics, OEM testing, backup-restore drill	Launch criteria §3.4 except closed test; load-test baseline (§14.4)	1–2 wk
M9 Closed testing	Play closed track, ≥12 testers for 14 days, declarations submitted, fixes	Production access granted	≥ 2 wk
M10 Pilot launch	Production release in pilot areas; monitoring; moderation rota	Success metrics §3.5 tracked	—

17. Development Workflow with Claude Code

17.1 Tools (installed once by Rahul)

Tool	Purpose / setup
Android Studio + Claude Code plugin (JetBrains Marketplace)	IDE; Claude Code diffs shown in the IDE.
Claude Code CLI	Run <code>claude</code> in Android Studio's terminal at the repo root.
git + GitHub CLI (gh)	<code>gh auth login</code> once; Claude Code uses <code>gh</code> to create the repo and PRs.
Node.js LTS + pnpm; JDK 17 (bundled with Android Studio)	Backend and Android builds.
Google Cloud SDK (gcloud)	Only for setup tasks Rahul runs himself; Claude Code gets no production credentials.
Notion MCP server	<code>claude mcp add --transport http notion https://mcp.notion.com/mcp</code> , then run <code>/mcp</code> inside Claude Code and complete the OAuth login.
Excalidraw tooling	Repo script <code>tools/diagrams</code> (created in P001) turns a small JSON description into a <code>.excalidraw</code> file plus SVG/PNG exports; <code>.excalidraw</code> files open and edit at <code>excalidraw.com</code> .

17.2 Repository governance files (created by P001)

File	Content
CLAUDE.md	Project instructions Claude Code loads every session: summary, repo map, golden rules (below), conventions, quality-gate commands, documentation duties, SOS rules, pointer to <code>docs/plan/</code> .
<code>.claude/settings.json</code>	Permission deny rules: force-push, merging PRs, reading secret files. The deny rule for pushing to main lives in <code>.claude/settings.local.json</code> (git-ignored), added at the end of P001 so P001's own pushes to main are not blocked.
<code>.claude/commands/start-prompt.md</code>	Slash command <code>/start-prompt</code> : the preflight in §17.4.
<code>.claude/commands/ship-prompt.md</code>	Slash command <code>/ship-prompt</code> : quality gate → commit → push → PR → repo log → Notion → report (§17.5).
<code>.githubhooks/pre-push</code>	Rejects any push to refs/heads/main (defence in depth); enabled with git config <code>core.hooksPath .githubhooks</code> .
<code>.github/pull_request_template.md</code>	PR body sections (§17.6).
<code>.gitmessage</code>	Commit message template (§17.6).
<code>docs/plan/</code>	This PDF + diagram sources, so Claude Code can consult the plan.
<code>docs/prompt-logs/</code>	One markdown log per prompt — the source-controlled copy of the Notion page.
<code>docs/adr/</code> , <code>docs/diagrams/</code> , <code>docs/runbooks/</code>	ADRs, Excalidraw sources + exports, operational runbooks.

CLAUDE.md – golden rules (verbatim)

1. One prompt = one branch = one pull request. Only P001 commits to main.
2. Never push to main. Never merge, approve or close a pull request. Rahul merges on GitHub.
3. Start every prompt with `/start-prompt`; finish with `/ship-prompt`.
4. Implement only the prompt's scope. Out-of-scope problems become Notion follow-ups.
5. Never commit secrets, keys, keystores, `.env` files or `google-services.json`.
6. Never push code that fails the quality gate. If you cannot make it pass, stop and report.
7. Document every prompt: `docs/prompt-logs/NNN-slug.md` + Notion Prompt Log page.
8. Draw a diagram in Excalidraw whenever a flow, state machine, schema or infrastructure changes.
9. SOS / live-location code requires the failure-matrix tests from Plan v7 §7.5.
10. Explain Android concepts you use in the 'Learning notes' section for Rahul.

17.3 Branch, commit and PR naming

Item	Convention	Example
Branch	<type>/<NNN>-<core-work>; type ∈ feat, fix, chore, ci, docs, test	feat/014-sos-device-flow
Commit title	Conventional Commits + prompt tag	feat(sos): device-first SOS trigger with offline SMS [P014]
PR title	Same as the main commit title	feat(sos): device-first SOS trigger with offline SMS [P014]
Merge method	Squash and merge (one commit per prompt on main)	—

17.4 Preflight (/start-prompt)

- `git switch main && git pull --ff-only`; confirm a clean working tree.
- Confirm the previous prompt's PR is merged (`gh pr list --state merged`). If not merged, stop and ask Rahul — do not stack branches unless the prompt says so.
- Update the previous prompt's Notion page: Status = Merged, merge commit SHA, merge date.
- Read CLAUDE.md, the prompt's plan sections in docs/plan/, related ADRs and the previous prompt log.
- Create the branch: `git switch -c <type>/<NNN>-<core-work>`.
- Create the Notion page for this prompt with Status = In progress, and post a short plan in the terminal.

17.5 Finish (/ship-prompt)

- Run the quality gate for every area touched (backend: typecheck, lint, test, build; contracts: regenerate openapi.json, diff must be committed, breaking-change check; Android: `./gradlew lint testDebugUnitTest assembleDebug`). Fix failures; never skip tests.
- Draw/update the Excalidraw diagram if §17.7 requires it; export SVG/PNG into docs/diagrams/.
- Write docs/prompt-logs/NNN-core-work.md using the Notion page template (§17.8).
- Commit with a detailed message (template below); push the branch (`git push -u origin <branch>`).
- Open the PR with `gh pr create` using the PR template; link the Notion page.
- Complete the Notion page (Status = In review, PR URL, commit SHA, test results, diagram link).
- Final terminal report: branch, commit SHA, PR URL, Notion URL, quality-gate results, follow-ups, and the manual review checklist for Rahul.

Commit message template (.gitmessage)

feat(sos): device-first SOS trigger with offline SMS [P014]

Why:

- Plan v7 §7.1–7.5: SOS must alert contacts without server or mobile data.

What changed:

- SosForegroundService (type=location), 5 s fixes stored in Room
 - SmsManager path + ACTION_SENDTO fallback; 112 dialer screen
 - SosSyncWorker (expedited, backoff) posting clientSosId

How tested:

- `./gradlew testDebugUnitTest` (42 passed) · lint clean · assembleDebug OK
 - Failure matrix rows 1,2,4,5,6,7,8 verified (see prompt log)

Risks / follow-ups:

- SEND_SMS Play declaration pending; intent fallback active

Refs: Plan v7 §7 · docs/prompt-logs/014-sos-device-flow.md · Notion: <url>

17.6 Pull request template

```
## Summary                (what and why, 3-5 lines)
## Plan references        (Plan v7 sections)
## Changes                (by area: android / backend / contracts / infra / docs)
## API contract           (openapi.json diff summary; breaking? yes/no)
## Database               (migrations; expand/contract step)
## Tests & quality gate   (commands + results)
## Diagrams              (docs/diagrams/... links)
## Security & privacy     (new data, permissions, secrets check)
## How Rahul can verify   (step-by-step manual checks)
## Follow-ups             (links to Notion items)
## Notion page            (URL)
```

17.7 Excalidraw diagrams

- Required when a prompt adds or changes: a user flow, a state machine, a data model/schema, an API interaction sequence, infrastructure/deployment, or the CI pipeline. Otherwise the Notion page states 'No diagram needed'.
- Files: docs/diagrams/NNN-<topic>.excalidraw + .svg + .png, plus the JSON description used to generate it, so the diagram can be regenerated.
- Keep diagrams under ~20 nodes; colour by role (blue = app/UI, green = services, orange = data, red = security/emergency, purple = infra, grey = external).
- The Notion page links the SVG on GitHub and attaches the PNG where the Notion MCP supports file upload.
- Starter diagrams shipped with this plan (committed by P001): 01-prompt-workflow, 02-mvp-architecture, 03-sos-device-first, 04-scaling-stages.

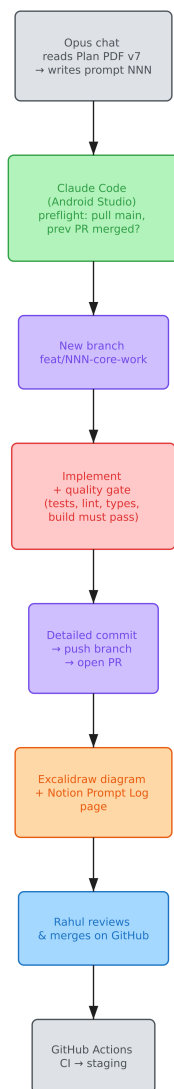
17.8 Notion documentation

Workspace structure (created by P001 under a parent page Rahul provides): **SafeRoute Kolkata — Engineering** containing a **Prompt Log** database, an **Architecture Decisions** database, a **Runbooks** page and a **Follow-ups** database.

Prompt Log property	Type / values
Prompt	Title, e.g. 'P014 · SOS device flow'
Number	Number
Status	Select: Planned · In progress · In review · Changes requested · Merged
Milestone	Select: M0 ... M10
Branch · PR URL · Commit SHA · Merge SHA	Text / URL
Quality gate	Select: Pass · Fail
Diagram	URL (GitHub SVG)
Date started · Date merged	Date

- Page template (every prompt):** 1 Objective · 2 Context & prerequisites · 3 Workflow executed (step-by-step, including commands) · 4 Changes (files, API contract diff, migrations) · 5 Diagram · 6 Quality gate & test results (commands, pass/fail counts, failure-matrix rows) · 7 Decisions & ADRs · 8 Security & privacy notes · 9 Known issues & risks · 10 Follow-ups & prerequisites for the next prompt · 11 How Rahul can verify · 12 Learning notes (Android/Kotlin concepts explained).
- If the Notion MCP is unavailable, Claude Code still writes docs/prompt-logs/NNN-*.md, reports the failure, and the next session syncs it to Notion during preflight.
 - Rahul's review adds comments on the PR; if changes are requested, Claude Code works on the same branch, appends a 'Revision' section to the log and Notion page, and pushes new commits.

17.9 Lifecycle and Rahul's review checklist



Rahul's review before merging

- PR title/branch follow §17.3; one prompt only.
- CI green; quality-gate results pasted in the PR.
- Changes match the prompt's scope and non-goals.
- No secrets, keys, keystores or google-services.json in the diff.
- openapi.json diff understood; breaking changes have an ADR.
- Migrations are expand-only unless the prompt says contract.
- For SOS/live location: failure-matrix rows listed and passed.
- Diagram present when required; Notion page complete.
- Run the 'How Rahul can verify' steps on a device.
- Squash and merge; delete the branch.

18. Prompt Catalog (P001-P022)

Every prompt after P001 uses its own branch and PR. 'Dgm' = Excalidraw diagram required (§17.7).

#	Branch / core work	MS	Depends	Core deliverables	Dgm
P001	main (direct) — repo bootstrap	M0	—	Monorepo, governance files, plan + diagrams committed, GitHub repo created, Notion workspace structure	Yes
P002	feat/002-backend-skeleton	M1	P001	Hono app, config, structured logs with request_id, /health, tests	No
P003	feat/003-db-foundation	M1	P002	Drizzle, PostGIS, first migrations, test container	Yes
P004	feat/004-openapi-contract-pipeline	M1	P002	Zod → OpenAPI, committed spec, breaking-change diff in CI	Yes
P005	feat/005-firebase-auth-api	M1	P003,P004	ID-token middleware, /v1/me bootstrap/get, roles	Yes
P006	ci/006-gcp-staging-deploy	M1	P005	WIF, Artifact Registry, Cloud Run staging, migration job	Yes

#	Branch / core work	MS	Depends	Core deliverables	Dgm
P007	feat/007-android-app-shell	M2	P001	Compose shell, Hilt, navigation, design tokens (Rahul creates the project with the Empty Compose wizard first)	Yes
P008	feat/008-android-api-client	M2	P004,P007	Generated Kotlin client, auth interceptor	No
P009	feat/009-signin-onboarding	M2	P005,P008	Firebase phone sign-in, consent screens EN/BN	Yes
P010	feat/010-map-location	M3	P007	MapLibre + MapTiler, permission flow, current location	No
P011	feat/011-place-search	M4	P005,P010	Geocoding adapter + endpoint, search UI	No
P012	feat/012-osrm-routing	M4	P011	OSRM image, routing adapter, alternatives UI, follow-me	Yes
P013	feat/013-emergency-contacts	M5	P009	Contacts API + UI, invite, opt-out	No
P014	feat/014-sos-device-flow	M5	P013	Arm/countdown, FGS, SMS paths, 112, Room, recovery	Yes
P015	feat/015-sos-server-orchestration	M6	P014,P006	SOS API, idempotency, state machine, outbox, pg-boss, FCM worker	Yes
P016	feat/016-live-location-share	M6	P015	Share sessions, batches, viewer page, revocation	Yes
P017	feat/017-incident-reports	M7	P010,P005	Report API + UI, collection_phase, abuse controls	Yes
P018	feat/018-moderation-web-export	M7	P017	Moderator app, decisions, audit log, CSV/XLSX export	Yes
P019	feat/019-safety-layer-exposure	M7	P012,P018	H3 aggregation, cells endpoint, overlay, metric v1	Yes
P020	feat/020-privacy-retention-backups	M8	P019	Deletion/export, purge jobs, backup export + drill runbook	Yes
P021	test/021-load-test-baseline	M8	P020	k6 scenarios A-C, capacity dashboard, SLO report	Yes
P022	chore/022-release-readiness	M9	P021	Signed AAB pipeline, Play track upload, launch checklist	Yes

19. Prompt Template & Worked Example (P001)

19.1 Template (fill every field)

PROMPT P<NNN> – <Core work title>
 Branch: <type>/<NNN>-<core-work> (P001 only: main)
 Commit / PR title: <type>(<scope>): <summary> [P<NNN>]
 Milestone: M<n> Depends on: P<...> Plan refs: v7 §<...>

1. Objective	6. Architecture constraints	11. Security & privacy
2. Context (what exists)	7. API / DB contract changes	12. Observability
3. Prerequisites & inputs Rahul must provide	8. Files / components	13. Documentation (log, Notion, ADR)
4. Non-goals	9. Implementation steps	14. Diagram (required? topic)
5. Requirements	10. Tests & failure cases	15. Definition of Done
		16. Git workflow & final report

Start with /start-prompt. Finish with /ship-prompt. Never push to main or merge.

19.2 Worked example — P001 repository bootstrap

PROMPT P001 – Repository bootstrap

Branch: main (the only prompt that commits directly to main)

Commit title: chore(repo): bootstrap SafeRoute monorepo and governance [P001]

Milestone: M0 Depends on: — Plan refs: v7 §0, §4, §17, §18

1. Objective: create the SafeRoute Kolkata monorepo with governance files, commit the plan and starter diagrams, create the private GitHub repo, push main, and set up the Notion engineering workspace.
3. Inputs Rahul must provide: <GITHUB_OWNER>, repo name (saferoute-kolkata), <NOTION_PARENT_URL>, local path of Plan v7 PDF and the diagrams folder.
Rahul has already run: gh auth login; claude mcp add ... notion; /mcp login.
4. Non-goals: no backend code, no Android project, no CI deployment.
9. Steps:
 - a. Create: android/ backend/ moderation/ contracts/ infra/ tools/diagrams/ docs/{plan,adr,diagrams,prompt-logs,runbooks} .github/workflows .claude/commands .githubhooks (each empty dir gets a README.md explaining its purpose).
 - b. Write CLAUDE.md with the golden rules (Plan §17.2) and quality-gate commands.
 - c. Write .claude/settings.json deny rules for: force push, gh pr merge, reading .env / *.jks / google-services.json (main-push deny comes in step l).
 - d. Write .claude/commands/start-prompt.md and ship-prompt.md (Plan §17.4, §17.5).
 - e. Write .githubhooks/pre-push rejecting refs/heads/main; .gitmessage; PR template; .gitignore (Android, Node, secrets); .editorconfig; README.md.
 - f. tools/diagrams: script turning JSON (nodes/edges/groups) into .excalidraw + SVG/PNG.
 - g. Copy Plan v7 PDF to docs/plan/ and starter diagrams to docs/diagrams/.
 - h. docs/adr/0001-kotlin-native-openapi.md and docs/adr/template.md.
 - i. .github/workflows/repo-checks.yml: markdown/JSON lint + secret scan (e.g. gitleaks).
 - j. git init -b main; commit with a detailed message;
gh repo create <GITHUB_OWNER>/saferoute-kolkata --private --source=. --remote=origin --push
 - k. Write docs/prompt-logs/001-repo-bootstrap.md (with the first commit SHA); commit and push it to main. This is the second and final direct commit to main.
 - l. Only now lock main locally: add a deny rule for pushes to main in .claude/settings.local.json (git-ignored) and run git config core.hooksPath .githubhooks; verify the hook rejects main:
echo "refs/heads/main 0 refs/heads/main 0" | .githubhooks/pre-push origin x -> exit 1
 - m. Try to enable a branch ruleset on main (require PR, require checks). If the GitHub plan does not support it for private repos, report that; the hook + CLAUDE.md rules apply.
 - n. Notion: under <NOTION_PARENT_URL> create 'SafeRoute Kolkata – Engineering' with the Prompt Log, Architecture Decisions and Follow-ups databases (Plan §17.8) and a Runbooks page; create the P001 page (Status = Merged, both commit SHAs) with the full template.
10. Tests: repo-checks workflow green on GitHub; hook test exits 1; tools/diagrams regenerates 01-prompt-workflow from its JSON without errors.
15. Done when: GitHub repo exists with green checks; Notion structure exists; report lists repo URL, commit SHAs, Notion URLs, and anything Rahul must do manually.

20. Definition of Done (every prompt)

- Scope matches the prompt; non-goals untouched; out-of-scope issues recorded as follow-ups.
- Quality gate passes for every touched area; CI green on the PR.
- OpenAPI contract and generated Kotlin client in sync; breaking changes have an ADR.
- No secrets in the diff or container image.
- Security/privacy and abuse controls reviewed; new personal data listed in the PR.
- Emergency side effects idempotent and auditable; failure-matrix rows tested where relevant.
- Migrations follow expand → migrate → contract; data continuity rules (§15) respected.
- Excalidraw diagram committed when required.
- docs/prompt-logs entry and Notion page complete, including follow-ups and learning notes.
- Branch pushed and PR opened — not merged. Rahul reviews and merges.

21. Post-MVP Roadmap

Stage	Items (from v5 phases, re-ordered)
P1 Reliability & reach	Quick Settings tile / widget SOS trigger; duress PIN; Hindi UI; voice turn-by-turn; offline map region
P2 Alert channels	TRAI DLT registration → server SMS; WhatsApp Business (verified templates); evaluate automated calls with legal review
P3 Official data	RTI aggregates ingested; jurisdiction polygons; Kolkata Police partnership; provenance dashboard
P4 Scale	SSE/WebSocket gateway; Redis for rate limits/caching; read replicas; precomputed route metrics; Cloud Armor
P5 Platforms	iOS (Kotlin Multiplatform for shared domain logic, or native Swift); Wear OS SOS
P6 Expansion	Additional cities only after moderation capacity and data partnerships exist

22. Risk Register

Risk / question	Impact	Mitigation / owner action
SEND_SMS declaration rejected	SOS needs one extra tap	Intent fallback built first; apply early in closed test.
OEM battery killers stop FGS	Tracking gaps	Battery-optimisation prompt, OEM guides, device matrix testing.
Sparse report data at launch	Overlay looks empty	Pilot areas, 'beta' label, OSM context layers.
False or malicious reports	Misleading map, defamation	Moderation, k-threshold, trust weights, delays.
Official data unavailable	Community data only	RTI requests early (M0); partnership outreach after M6.
Moderation capacity (solo)	Backlog, stale data	Cap daily reports, recruit backup moderators before launch.
Cloud cost (min instances, Cloud SQL)	Monthly spend	Smallest tiers, budget alerts, review after 30 days.
Police-station boundary data	Official layer blocked	Digitise from published maps; verify with police.
Legal review of SOS messaging	Liability	Clear 'not an emergency service' copy; legal review before P2.
Claude Code pushes to main or merges	Unreviewed code on main	CLAUDE.md rules, settings.json deny rules, pre-push hook, ruleset where available.
Notion MCP unavailable	Missing documentation	Repo prompt log always written; synced at next preflight.
Prompt too large to review	Bugs slip through	Split into NNNa/NNNb (\$0); ~800-line guideline.
Test data mixed with beta data	Polluted dataset	collection_phase tag; exclude closed_test in exports.
Puja-night overload	Stale live location	Stage-2 triggers, Scenario B load test before festival season.

23. References

- Google Play target API level requirements — developer.android.com/google/play/requirements/target-sdk
- Google Play testing requirements for new personal developer accounts — support.google.com/googleplay/android-developer/answer/14151465

- Google Play SMS and Call Log permissions policy — support.google.com/googleplay/android-developer/answer/9047303
- Claude Code documentation — docs.claude.com/en/docs/claude-code/overview
- Notion MCP (Claude Code setup) — developers.notion.com/docs/get-started-with-mcp
- Workload Identity Federation with deployment pipelines — cloud.google.com/iam/docs/workload-identity-federation-with-deployment-pipelines
- google-github-actions/auth — github.com/google-github-actions/auth
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- Youth Bar Association of India v. Union of India (2016) — online FIR publication and sensitive-offence exemption
- MapLibre — maplibre.org · OSRM — project-osrm.org · H3 — h3geo.org · Hono — hono.dev · pg-boss — github.com/timgit/pg-boss · k6 — k6.io · Protomaps — protomaps.com · Excalidraw — excalidraw.com

Document status: v7 — supersedes v6. Capacity figures are planning assumptions, not measurements; the GCP topology is an implementation target, not a claim of tested capacity. Legal and policy points are summaries for planning, not legal advice; confirm with current official sources and, for SOS messaging and data publication, a qualified lawyer.